

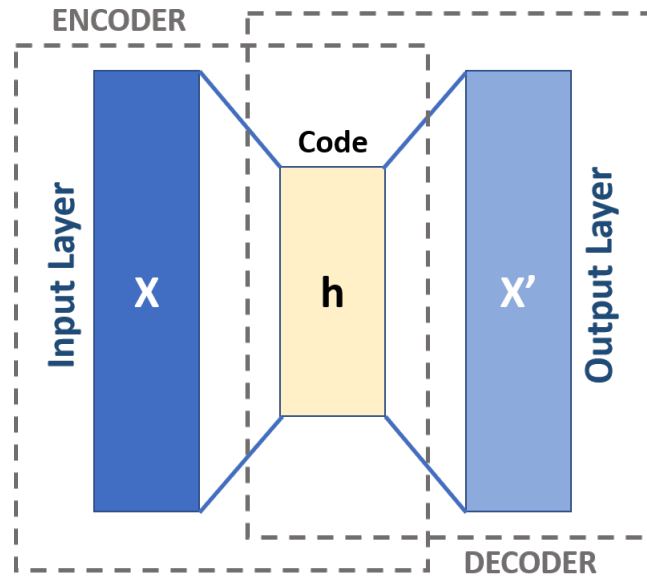
AUTOENCODERS & GENERATIVE MODELS

AI506 — Advanced Machine Learning

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- The main autoencoder **objective**
- **Types** of autoencoders (**under/over-complete** AE, **DAE**, **SAE**, **VAE**)
- **Challenges** of generative modeling, **likelihood** of the data
- **GANs**: **generator/discriminator** objective



Let f_θ be the encoder and g_ϕ the decoder, then

$$\arg \min_{\theta, \phi} \mathcal{L} = \|x - g_\phi(f_\theta(x))\|$$

$$f_{\theta} : \mathbb{R}^d \rightarrow \mathbb{R}^k$$

$$k < d$$

undercomplete

$$k \geq d$$

overcomplete (trivial)

Corrupt input x with random noise

$$\tilde{x} = x + \varepsilon \quad \text{where } \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

Now

$$\arg \min_{\theta, \phi} \mathcal{L} = \|x - g_{\phi}(f_{\theta}(\tilde{x}))\|$$

Recall L1 regularization

$$\|z\|_1 = \sum_i |z_i|$$

Let $z = f_\theta(x)$

$$\arg \min_{\theta, \phi} \mathcal{L} = \|x - g_\phi(z)\| + \lambda \|z\|_1$$

Instead of the regular encoder let

$$f : \mathbb{R}^d \rightarrow (\mathbb{R}^k, \mathbb{R}^k)$$

where $\mu \in \mathbb{R}^k$ and $\sigma^2 \in \mathbb{R}^k$ Now

$$z \sim \mathcal{N}(\mu, \sigma^2)$$

The decoder:

$$\hat{x} = g_{\phi}(z) \quad \text{where } z \sim \mathcal{N}(\mu, \sigma^2)$$

When generating $z \sim \mathcal{N}(0, I)$ We use

$$\arg \min_{\theta, \phi} \mathcal{L} = \|x - g_{\phi}(z)\| + \underbrace{D_{\text{KL}}(\mathcal{N}(\mu, \sigma^2) \parallel \mathcal{N}(0, I))}_{\text{KL Divergence}}$$

We cant backpropagate through the sampling

$$z = \mu + \sigma \cdot \varepsilon, \quad \text{where } \varepsilon \sim \mathcal{N}(0, 1)$$

One could also compute $\log p(x)$

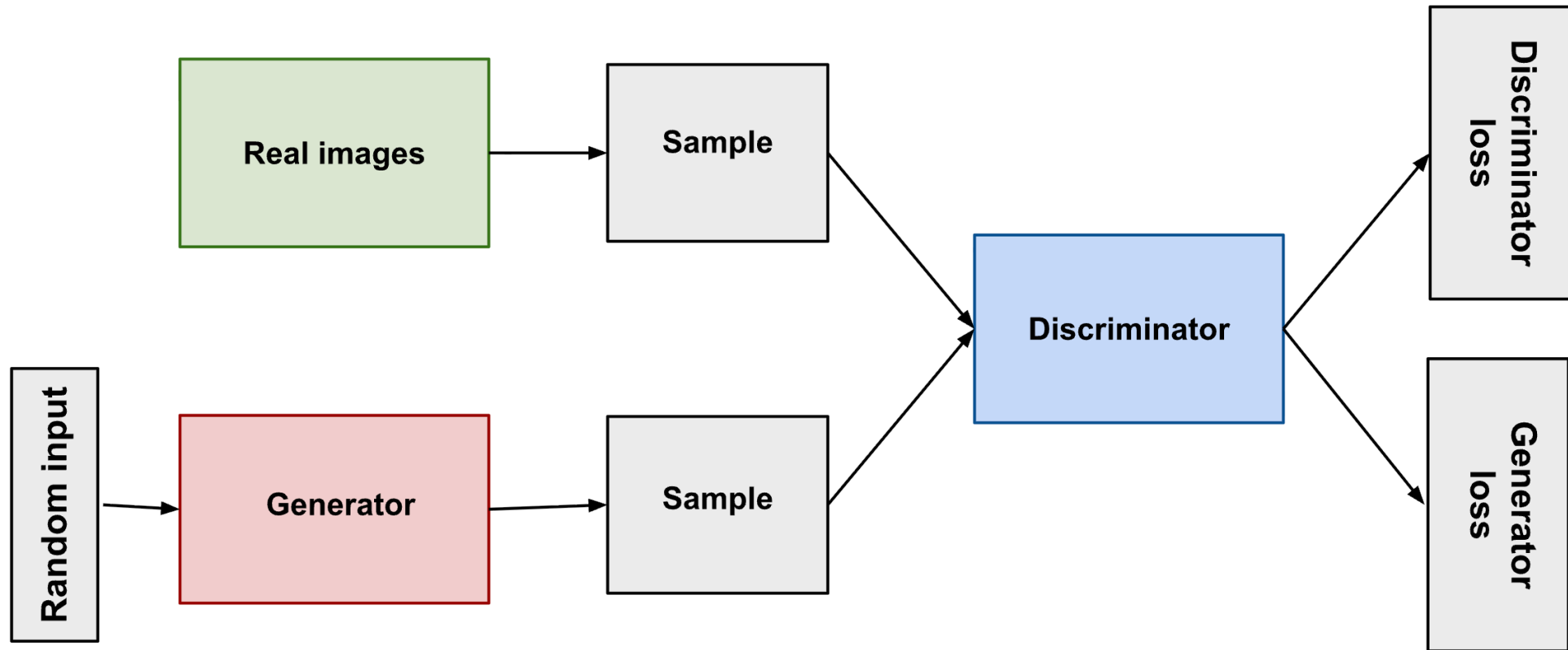
$$\log p(x) = \int_{\mathbb{R}^k} p(x|z)p(z) \, dz, \quad \text{where } \mathbb{R}^k \text{ is uncountably infinite}$$

Maximise a lower bound instead:

$$\log p(x) \geq \underbrace{\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]}_{\text{reconstruction}} - \underbrace{D_{\text{KL}}(q_\phi(z|x) \parallel p(z))}_{\text{KL divergence}}$$

where:

- $q_\phi(z|x) = \mathcal{N}(\mu, \sigma^2)$
- $p_\theta(x|z)$
- $p(z) = \mathcal{N}(0, I)$



Let $D_\phi : \mathbb{R}^d \rightarrow [0, 1]$

Let $G_\theta(z) \in \mathbb{R}^d$, where $z \sim \mathcal{N}(0, I)$.

Then the objective

$$\min_{\theta} \max_{\phi} (G_\theta, D_\phi) = \mathbb{E}_{x \in p_{\text{data}}} [\log D_\phi(X)] + \mathbb{E}_{z \in p(z)} [\log(1 - D_\phi(G_\theta(z)))]$$